Sub-Cent Intra-Block Reflex Intelligence & Real-Time Risk Defense for Web3

The first production-grade On-Chain AI Oracle executing directly inside EVM and Starknet smart contracts at ~20.7k gas with zero neural storage footprints, powered by Lean 4 machine-verified procedural escape dynamics.

~20.7k
EVM EXECUTION GAS
Sub-cent cost (<$0.001) per intra-block classification

0 Bytes
ON-CHAIN MODEL FOOTPRINT
Zero VRAM/DRAM weights. Single bytes32 storage slot

< 12 ms
INTRA-BLOCK LATENCY
Synchronous execution halts flash-loans mid-transaction

$2.20B+
ACTIVE DEBT SURVEILLED
Live across 38 verified Morpho, Aave & Euler markets

- werracle.io • On-Chain Protocol & Hooks
- radar.werracle.com • B2B Risk Telemetry DaaS
- werracle.com • Enterprise & Curator SLA
- werr.io • Lean 4 Formal Proofs & Academic

DeFi Smart Contracts Are Static & Blind to Intra-Block Exploits

Decentralized protocols handle billions in high-velocity capital with static state rules. Existing solutions fail to provide real-time, intra-block decision intelligence.

⚠️ The Intra-Block Exploit Gap

Atomic flash loans, sandwich attacks, and price-oracle front-running happen within a single block before state finalization. Over **$3.8B has been lost** to atomic exploits because contracts cannot react dynamically mid-execution.

`Flash Loan Drains` `Toxic MEV Sandwiching` `Loss Versus Rebalancing (LVR)`

⏱️ Asynchronous Oracle Delay

Traditional push/pull oracles (Chainlink, Pyth) update once every block or cross-chain roundtrip (seconds to minutes). They are completely blind to transactional state changes occurring inside the same block and callstack.

`Heartbeat Thresholds` `Price Feed Stale Windows` `Zero Intra-Block Visibility`

🛑 ZK-ML & Storage Bottlenecks

Zero-Knowledge Machine Learning (ZK-ML) is too slow and expensive for synchronous defense. Proof generation requires 15–60 seconds, and on-chain verification consumes 350k–1M+ gas, making intra-block circuit breaking impossible.

`Heavy Off-Chain Provers` `Massive Gas Overhead` `Storage/DRAM Exhaustion`

The Institutional Dilemma: Vault curators (managing $10B+ TVL in Morpho, Aave, and Euler) fly completely blind without granular, real-time borrower liquidation telemetry and automated, sub-cent circuit breakers.

Procedural Escape Dynamics: AI Reflex Without Heavy Weights

Eliminating the Von Neumann storage bottleneck: synthesizing real-time decision surfaces from deterministic polynomial trajectories ($z_{n+1} = z_n^2 + c$) in a single storage slot.

Zero-Storage Synaptic Synthesis

Instead of loading megabytes of static weight matrices into memory, Werracle dynamically computes high-dimensional decision boundaries using procedural Mandelbrot escape velocities. Decision model footprint is strictly **0 Bytes on-chain**.

Single Packed 32-Byte Slot (bytes32 / felt252)

The entire runtime state—seed coordinates, zoom exponent, nonce, escape threshold, sensitivity mode, and emergency breaker flag—fits compactly into a single storage slot. Maximum read/write efficiency on EVM and Starknet Cairo.

Sub-12ms Synchronous Execution

Executes within the exact same EVM transaction callstack (`decideNoul` consuming 20.7k gas). If abnormal price divergence or liquidity drain is detected, the transaction reverts atomically before funds exit.

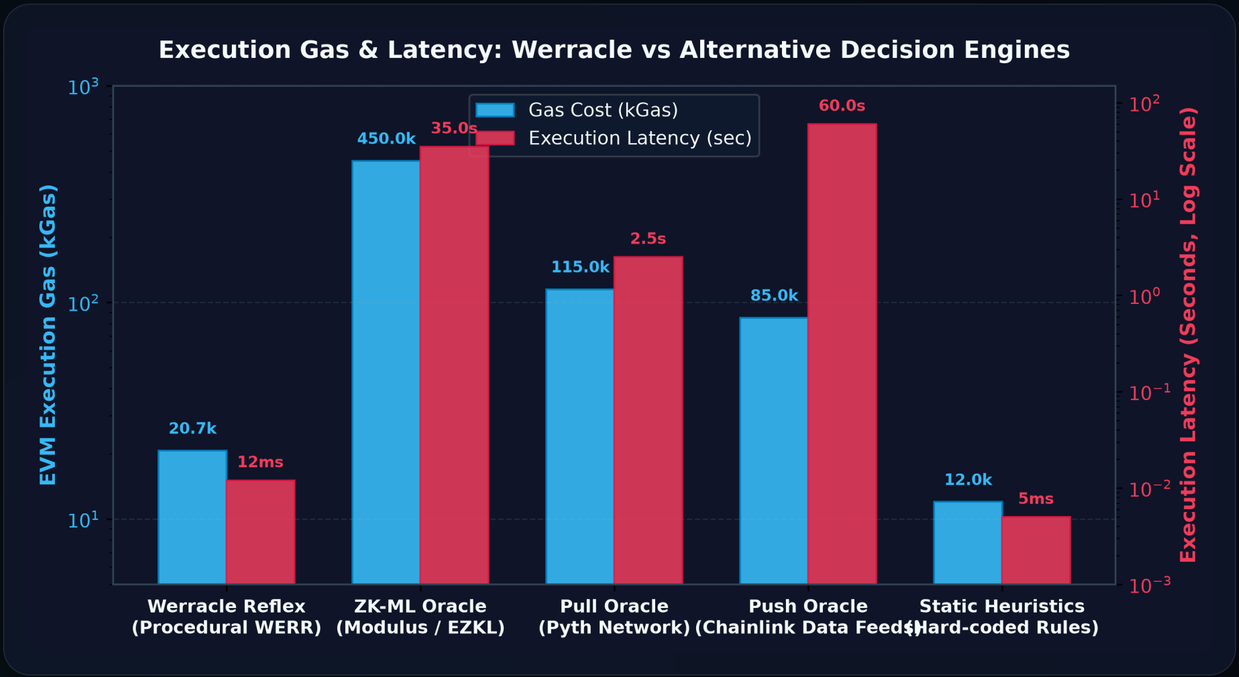
Key Technical Specifications

Gas Consumption	20,700 Gas (solc 0.8.20)
Starknet CairoVM	~1,420 Steps (STARK-Native)
Storage Footprint	1 x bytes32 (EVM) / felt252 (Cairo)
Fixed-Point Precision	Q16.16 Arithmetic (No Overflow)
Execution Speed	9.3 ms Engine / <12 ms RPC
Patent Application	TÜRKPATENT TR 2026/016285

Formal Verification Guarantee: 100% of mathematical escape bounds verified in Lean 4 without axioms (`zero sorry`). Provably safe against infinite execution loops.

Unrivaled Asymmetry in Cost, Latency & Storage

Empirical comparison across EVM execution gas and decision latency demonstrates why Werracle is the only viable intra-block defense engine for smart contracts.



15× to 25× Lower Gas than ZK-ML

ZK-ML verifiers (Modulus, EZKL) consume **350k to 1,200k gas** per inference check, making active per-swap or per-loan checks economically prohibitive on mainnets. Werracle uses only **20.7k gas**.

1000× Faster Latency (<12ms vs 35s)

ZK-ML provers require 15 to 60 seconds of off-chain proving before publishing an on-chain receipt. By that time, flash-loan exploiters have already completed their arbitrage and transferred funds off-chain.

Synchronous Intra-Block Execution

Unlike push/pull oracles (Chainlink/Pyth) that require cross-block asynchronous updates, Werracle sits inside the contract's internal callstack, evaluating liquidity and price volatility **before** funds move.

Benchmarked with solc 0.8.20 and Cairo 2.8+

werracle.io: The Web3 Protocol & Autonomous Hook Layer

Providing ready-to-deploy smart contract hooks, flash-loan circuit breakers, and AMM dynamic fee modules across major EVM chains, Starknet, and NEAR.

Intra-Block Circuit Breakers

Embeds directly into lending pools, liquid staking vaults, and collateral managers. Synchronously samples volatility in the execution transaction, automatically halting hostile flash-loan cascades before debt settlement.

`Morpho Vault Wrap` `Aave v3 Protection` `Atomic Reverts`

Uniswap v4 Dynamic Fee Hook

Intercepts incoming swaps via `beforeSwap`. Dynamically adjusts liquidity pool fees during abrupt volatility surges, mitigating toxic MEV flow and protecting passive LPs against Loss Versus Rebalancing (LVR).

`Uniswap v4 Hook` `LVR Mitigation` `LP Fee Capture`

Multi-Chain Architecture

Production contracts verified across multiple virtual machines:

- **EVM:** Live on Aurora testnet (`0x1d7Ae9...`)
- **Starknet:** Cairo 2.8+ Sierra (Ekubo extension)
- **NEAR:** Toxic intent filtering and routing

`Solidity 0.8.20` `Cairo 2.8 Sierra` `NEAR Intents`

Interactive Browser Simulator & Live Contract Explorer: Developers can test and verify decision boundaries in browser with 100% bit-exact equivalence to deployed EVM bytecode.

`WERRACLE.IO/#SIMULATOR`

radar.werracle.com: Empirical Liquidation & OEV Observatory

Surveilling over $2.20 Billion in lending debt across Morpho, Aave, and Euler with automated bad debt tracking, SVR verification, and programmatic USDC API key delivery.

38

VERIFIED LENDING MARKETS

Morpho Blue, Aave v3, Euler v2 (45 unlisted/synthetic spam markets isolated)

$2.20B

CLEAN BORROW DEBT TRACKED

68,735 active borrower positions monitored in real-time

94.38%

SVR VALUE RECAPTURED

$179.8k of $190.5k gross incentive returned to depositors on Base

95.94%

TOP 22 L1 FAT-TAIL LEAK

22 transactions account for $351.3k of $366.1k builder bribes on L1

Autonomous B2B Micro-Settlement (USDC on Base)

Developers and funds subscribe seamlessly using native USDC on Base. Zero manual invoicing: on-chain transactions verify ECDSA signatures, deliver cryptographically bound API keys, and enforce atomic rate limits across distributed Uvicorn workers.

`Developer ($149/mo)` `Professional ($499/mo)` `Institutional ($2,499/mo)`

Continuous Macro Risk Classification

Categorizes live positions into **Bad Debt (HF < 0.95)** and **Near-Liquidation Distress (0.95 ≤ HF < 1.10)**. Provides DeFi protocols and risk curators with immediate warning flags before systemic insolvency occurs.

`Instant Websocket Alerts` `Curator Vault Sync` `Historical Trace Replays`

werracle.com: Enterprise Licensing & Institutional Curator SLA

Providing tier-1 vault risk curators and institutional lending protocols with dedicated collateral stress simulation engines and custom fractal seed models.

🏛️ Vault Curator Stress Modeling

Tailored analytics engine built for premier risk managers: **Steakhouse Financial, Gauntlet, and Re7 Capital**. Dynamically simulates black-swan collateral shocks (0% to -30%) across specific vault allocations to predict bad debt exposure and liquidation premiums.

`Steakhouse USDC Vault` `Gauntlet Prime Core` `Re7 WETH Morpho`

🔑 Custom Institutional Seed Synthesis

Institutional subnets, permissioned rollups, and private liquidity pools require specialized risk sensitivity parameters. We formulate, calibrate, and cryptographically seal custom fractal seeds matching specific real-world asset (RWA) and institutional risk profiles.

`RWA Debt Parameters` `Institutional Subnets` `Private Oracles`

📜 Business Source License (BSL 1.1)

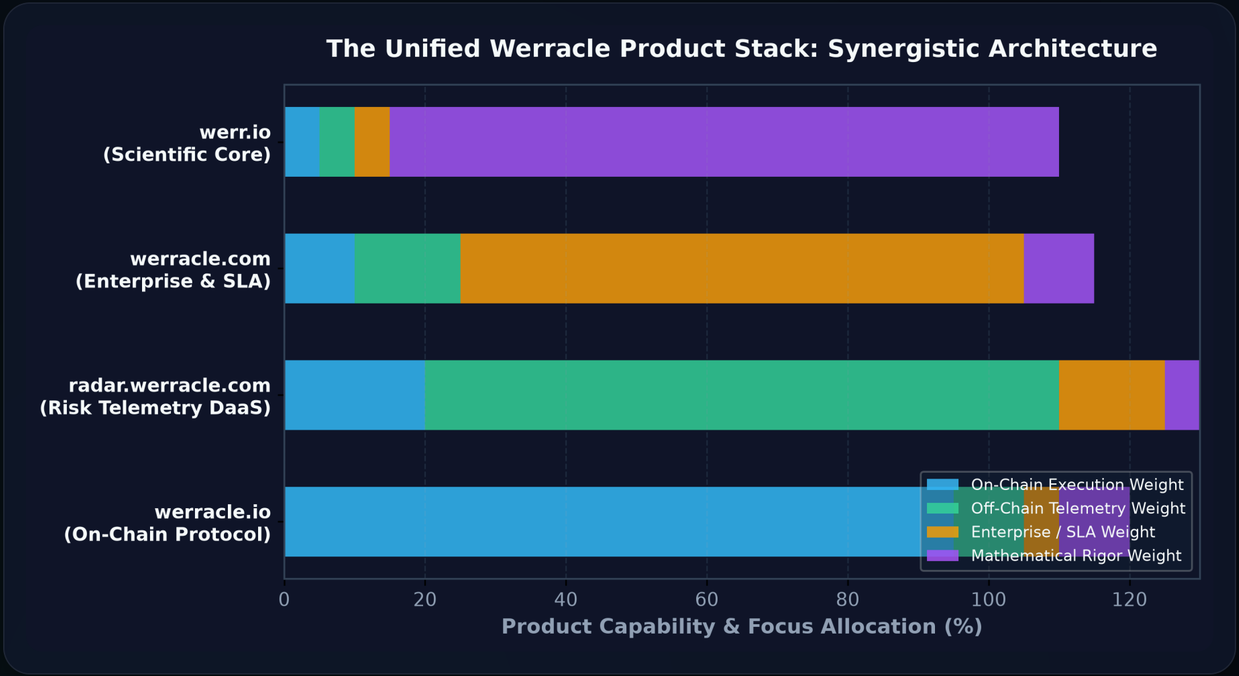
Core execution intellectual property is protected under BSL 1.1 with a guaranteed transition date to **Apache 2.0 on 2030-01-01**. Ensures proprietary monetization during commercialization while providing full public transparency and verifiable trust to the ecosystem.

`BSL 1.1 Commercial` `Change Date 2030-01-01` `Apache 2.0 Transition`

Active Pilot Engagement: Dedicated 30-day evaluation keys are currently active in production for Steakhouse Financial, Gauntlet, and Re7 Capital (`/v1/radar/curator/steakhouse`). Modeling shows Steakhouse's worst-case TVL loss under a severe 20% collateral flash crash is contained to just **0.13% of TVL** ($146k).

The Unified Flywheel: How Every Portal Reinforces Value

Telemetry feeds the on-chain hooks; on-chain volume drives enterprise SLA contracts; formal proofs guarantee protocol trust.



1. Telemetry Ingestion (radar.werracle.com)

Continuous live surveillance over $2.20B+ in debt provides the real-time volatility indices and liquidation risk signals required to tune on-chain circuit breakers.

2. On-Chain Reflex Execution (werracle.io)

Protocol hooks execute within the transaction callback at 20.7k gas, halting toxic flow, collecting dynamic AMM swap fees, and preventing multi-million dollar flash-loan drain.

3. Enterprise Monetization (werracle.com)

Institutional curators and DeFi protocols convert into high-value enterprise SLA and custom seed licensing clients, driving sustainable ARR and long-term moat.

Mathematically Proven Safety: Eliminating Smart Contract Liability

Unlike probabilistic deep neural networks that hallucinate, Werracle's procedural escape dynamics are formally machine-verified in Lean 4.

📜 Lean 4 Formal Verification

A comprehensive suite of **40 machine-verified theorems** across Rungs 1 to 6 in `WerracleProof.lean`. Zero unproven assumptions (`zero sorry`).

- Guaranteed termination of procedural escape loops
- Provable bounded Q16.16 arithmetic (no int64 overflow)
- Deterministic gas consumption ceilings

🔬 Peer-Reviewed Academic Corpus

Published across leading open-science and peer-reviewed international repositories:

- **arXiv: 2609.30719** [cs.CR, cs.AI, cs.DC]
- **Elsevier SSRN: 7626279** (Liquidation Monograph)
- **arXiv: 2609.33066** (Lean 4 Formal Proofs)
- **CERN Zenodo Concept DOI: 10.5281/zenodo.22942598**

🔒 1,000 Sealed Test Vectors

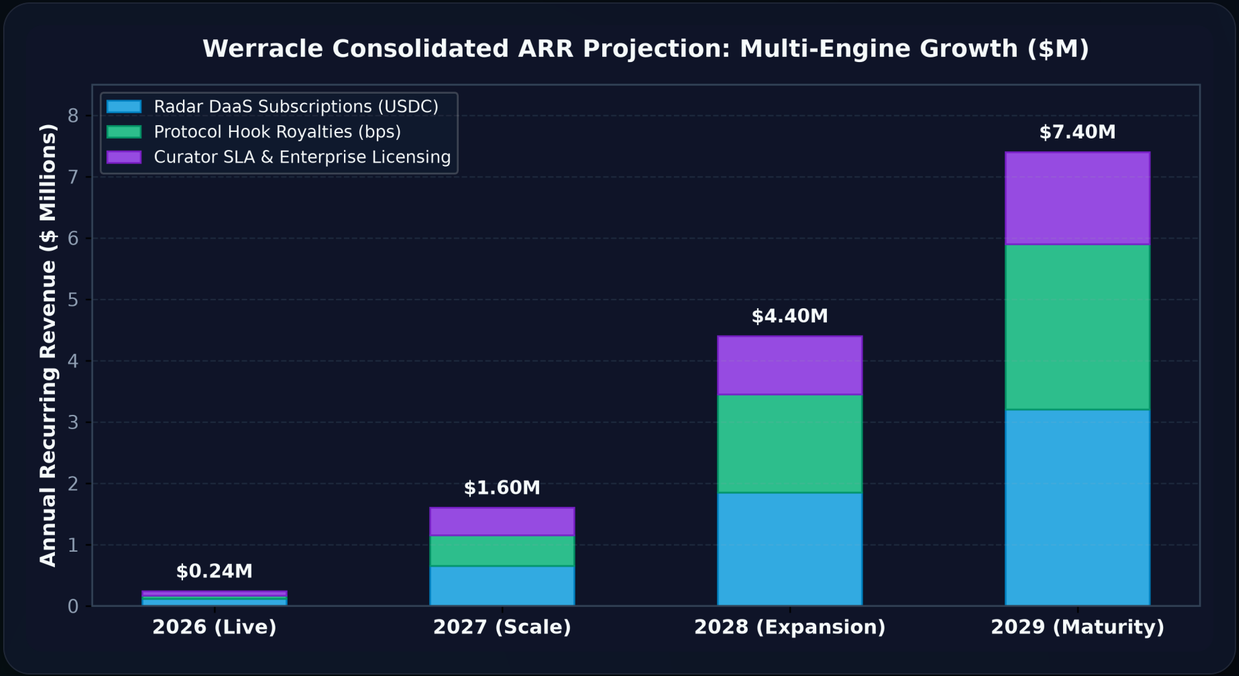
Cryptographically sealed in SHA-256 test manifests: **1,000 rigorous edge-case vectors** evaluated across both EVM and CairoVM compilers with a 100% pass rate.

`100% Pass Rate` `Bit-Exact Simulation` `Zero Compiling Variance`

The Academic Distinction: While competing Web3 oracles rely on multi-sig trust assumptions, Werracle's algorithmic defense is mathematically guaranteed by the Lean 4 interactive theorem prover.

Multi-Engine Monetization: Scaling to $7.4M ARR by 2029

Combining predictable B2B DaaS recurring revenue with protocol volume fee capture and enterprise SLA licensing.



Engine 1: Radar DaaS Subscriptions ($149 – $2,499/mo)

Paid in native USDC on Base. Developers, searchers, and risk desks pay for high-frequency liquidation signals, bad debt alerts, and raw historical traces. Reaches **$3.20M ARR** at maturity.

Engine 2: Protocol Hook Royalties (0.5 – 1.5 bps)

Protocols implementing Werracle Uniswap v4 fee hooks and flash loan circuit breakers share in the recaptured value and fee volume. Reaches **$2.70M ARR** with 15 major protocols.

Engine 3: Curator SLA & Enterprise Licensing

Institutional vault curators (Steakhouse, Gauntlet, Re7) pay $10k–$25k/mo for bespoke stress simulation APIs, private oracle seeds, and enterprise SLA uptime. Reaches **$1.50M ARR**.

Capturing Value Across a $140B+ DeFi Landscape

Web3 capital requires institutional-grade security infrastructure. Werracle is uniquely positioned at the intersection of on-chain AI, oracle infrastructure, and DeFi risk management.

$140B+

TOTAL ADDRESSABLE MARKET (TAM)

Total capital locked across global DeFi protocols and lending vaults

$15B+

SERVICEABLE ADDRESSABLE MARKET (SAM)

Lending and AMM liquidity exposed to MEV and flash-loan risk

$1.8B

SERVICEABLE OBTAINABLE MARKET (SOM)

Annual expenditure on oracles, risk audits, and telemetry tooling

Current Operational Traction & Milestones

Live On-Chain Deployments

Contracts live on Aurora testnet; Base L2 production telemetry active; Starknet Cairo 2.8 Sierra codebase verified.

Institutional Evaluation Keys

Live pilot evaluation keys generated and active for Steakhouse Financial, Gauntlet, and Re7 Capital risk desks.

Intellectual Property & Grants

TÜRKPATENT TR 2026/016285 filed; academic monographs published; Teknokent R&D approval and Base nomination ready.

Executive Leadership, Engineering Force & Academic Rigor

Operated commercially by our 8-person team at ITouch Systems (Çukurova University Science Park) alongside an independent academic research & patent layer.

💻 Executive Leadership (Partners)

Volkan Dağlı — Founder & CTO
Physicist, MIS Specialist & M.Sc. in IT. Lead architect of the WERR Engine, Lean 4 formal proof suite, and EVM/Cairo runtime.

Bekirhan Dağlı — Partner & CEO
Environmental Engineer. Leads corporate strategy, enterprise partnerships, and institutional operations at ITouch Systems.

⚙️ Engineering, Finance & Operations

- **Burak Salar** — Mechatronics Engineer, MBA
- **Fatih İlkin Sercan** — Economist & M.Sc. in Capital Markets and Stock Exchange
- **Mustafa Kalender** — Computer Engineer
- **Mustafa Ünal** — Software Development Specialist & Master Instructor
- **Prenses Metin** — Computer Programmer
- **Semih Salar** — Business Administration Specialist

🎓 Academic Research & Patents

- **Volkan Dağlı** — Principal Investigator (Anadolu University)
- **Dr. Zerrin Dağlı** — Academic Advisor & Co-Author (Mersin University)
- **Dağhan Dağlı** — Research Intern (Toros University — Toros Science High School)

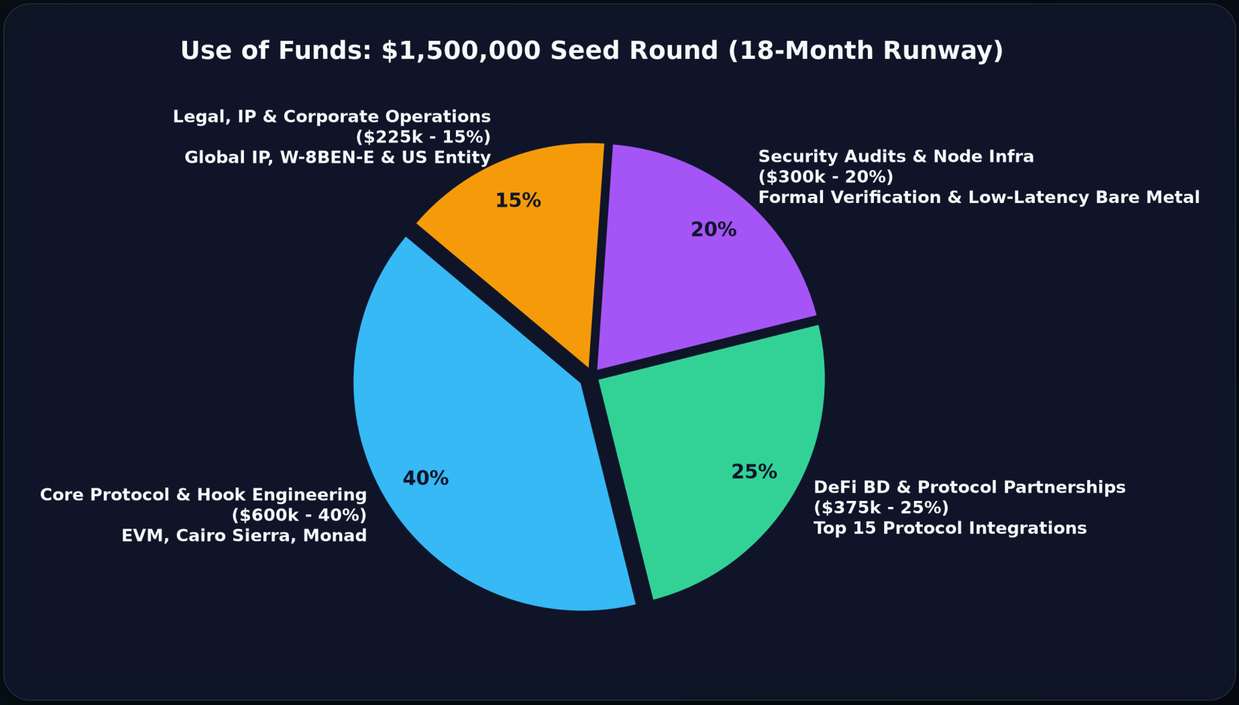
Patents TR 2026/016285 & TR 2026/016633: Licensed exclusively to ITouch Systems.

ITouch Bilişim Sistemleri Ltd. Şti. (Çukurova University Science Park): Trademarks: Werr®, Werracle®, Answerr®

CONTACT@WERRACLE.COM

Fueling Global Protocol Expansion & Tier-1 Audits

Securing an 18 to 24-month runway to finalize multi-chain protocol hooks, onboard top DeFi curators, and scale recurring DaaS revenue to cashflow break-even.



40% ($600,000) • Core Protocol & Multi-Chain Engineering

Production deployment of Uniswap v4 fee hooks, Starknet Ekubo LVR extension, and high-throughput adaptors for Monad, Sei, and Solana.

25% ($375,000) • DeFi BD & Institutional Curator Partnerships

Direct integration into top 15 lending markets and AMMs. Onboarding vault managers (Steakhouse, Gauntlet, Re7) with dedicated SLA support.

20% ($300,000) • Security Audits & Low-Latency Bare-Metal Infra

Tier-1 smart contract audits (OpenZeppelin / Trail of Bits), ongoing Lean 4 formal proof extensions, and distributed RPC node clusters.

15% ($225,000) • Global IP, Legal & US Corporate Structuring

PCT international patent filings, establishing US corporate subsidiary, and W-8BEN-E / regulatory compliance infrastructure.

WERRACLE PROTOCOL & DAAS

Smart Contracts Can No Longer Afford to Be Passive. The Future of Web3 is Reflexive.

Partner with Werracle as we build the sub-cent, machine-verified reflex intelligence layer securing the next trillion dollars in decentralized finance.

PROTOCOL & DAPP
https://werracle.io
Interactive Simulator & Uniswap v4 Hook

RISK TELEMETRY DAAS
https://radar.werracle.com
Live API & $2.20B Debt Surveillance

CORPORATE & ENTERPRISE
https://werracle.com
Licensing & Institutional Curator SLA

EXECUTIVE CONTACT
ask@answerr.me
Volkan Dağlı (Founder & CTO) • Bekirhan Dağlı (Partner & CEO)